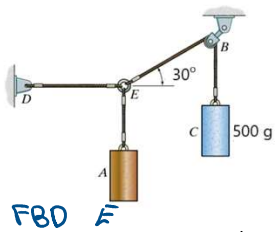


Exercise 1 (20%)

If the mass of cylinder C is 500 g, determine:

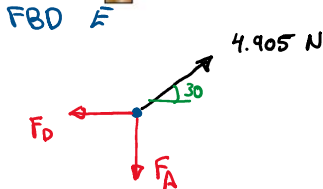
- a) The mass of cylinder A in order to hold the assembly in the position shown in the figure.
b) The tension in the cable DE.



$$m_c = 500g = 0.5 \text{ kg}$$

$$W_c = 0.5 \text{ [kg]} \cdot 9.81 \text{ [m/s}^2\text{]} = 4.905 \text{ [N]}$$

↓
Tension cable EBC



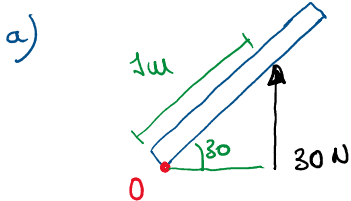
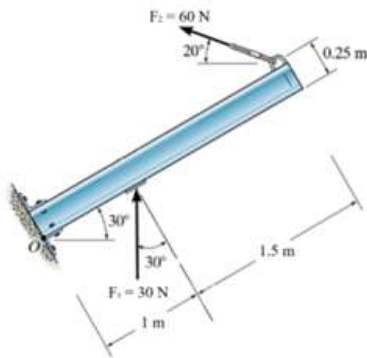
$$\begin{cases} +\uparrow \Sigma F_y = 0 \Rightarrow 4.905 \cdot \sin 30 - F_A = 0 \Rightarrow F_A = 2.4525 \text{ [N]} \\ \rightarrow \Sigma F_x = 0 \Rightarrow -F_D + 4.905 \cos 30 = 0 \Rightarrow F_D = 4.2478 \text{ [N]} \end{cases}$$

Tension cable DE $\rightarrow F_D = 4.25 \text{ N}$

Mass cylinder A $\Rightarrow F_A = m_A \cdot \vec{g} \Rightarrow m_A = \frac{2.4525 \text{ [N]} \cdot \text{kg} \cdot \frac{\text{m}}{\text{s}^2}}{9.81 \text{ [m/s}^2\text{]}} = \underline{\underline{0.25 \text{ kg}}}$

Exercise 2 (20%)

- A) Determine the moment of the force F1 acting on the beam shown in the figure about point O.
B) Determine the moment of the force F2 acting on the beam shown in the figure about point O.

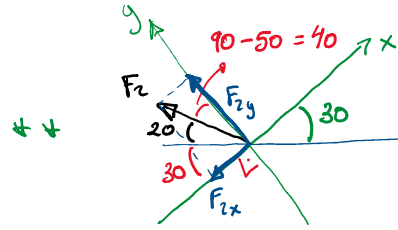
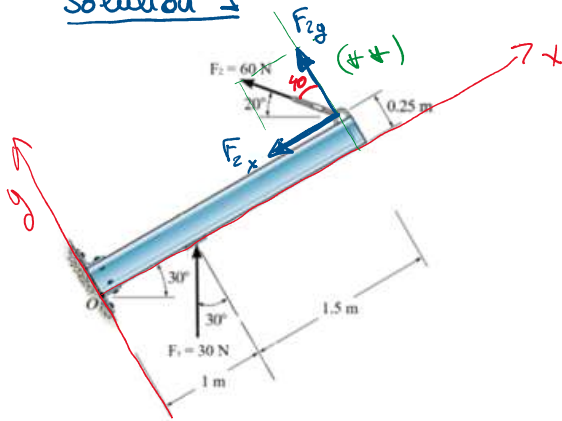


$$(\sum M_{F_1})_O = d \cdot F_1 = 1 \cdot \cos 30 \text{ [m]} \cdot 30 \text{ [N]} = \underline{\underline{25.98 \text{ Nm}}}$$

⤴

b)

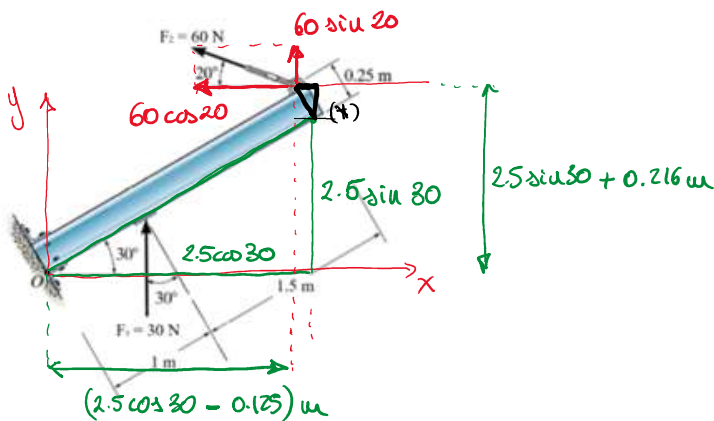
Solution I



$$\begin{cases} F_{2x} = F_2 \sin 40 = 60 \cdot \sin 40 = 38.567 \text{ N} \\ F_{2y} = F_2 \cos 40 = 60 \cdot \cos 40 = 45.963 \text{ N} \end{cases}$$

$$\sum (M_{F_2})_O = 0.25 \text{ [m]} \cdot 38.567 \text{ [N]} + 1.5 \text{ [m]} \cdot 45.963 \text{ [N]} = \underline{\underline{124.5 \text{ Nm}}}$$

Solution II

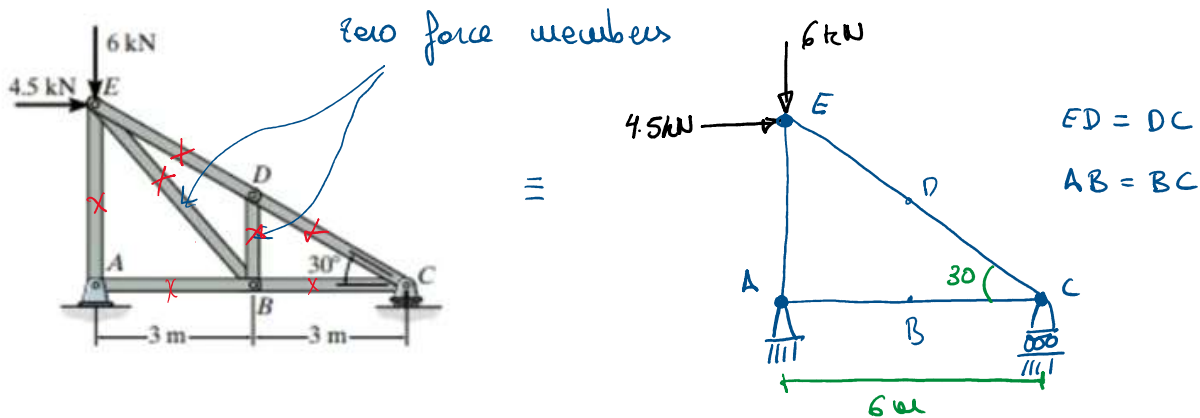


$$\begin{aligned} (*) \quad & 0.25 \cdot \sin 30 = 0.125 \text{ m} \\ & 0.25 \text{ m} \cos 30 = 0.216 \text{ m} \end{aligned}$$

$$\sum (M_{F_2})_O = (2.5 \sin 30 + 0.216) \text{ [m]} \cdot 60 \cos 20 \text{ [N]} + (2.5 \cos 30 - 0.125) \text{ [m]} \cdot 60 \sin 20 \text{ [N]} = \underline{\underline{124.5 \text{ Nm}}}$$

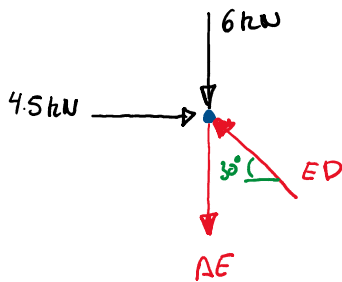
Exercise 3 (30%)

The truss in the figure has a fixed support (pin support) at point A and a roller support at point C. Determine the forces in all members and explain which of them are subjected to tensile or compressive forces.



Method of joints:

Joint E (FBD):



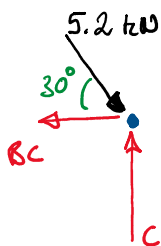
$$\rightarrow \sum F_x = 0 \Rightarrow 4.5 - ED \cos 30 = 0 \Rightarrow ED = 5.2 \text{ kN [C]}$$

$$\uparrow \sum F_y = 0 \Rightarrow -6 - AE + ED \sin 30 = 0 \Rightarrow AE = -3.4 \text{ kN}$$

↑
opposite direction

$$AE = 3.4 \text{ kN [C]}$$

Joint C (FBD):



$$\rightarrow \sum F_x = 0 \Rightarrow -BC + 5.2 \cos 30 = 0 \Rightarrow BC = 4.5 \text{ kN [T]}$$

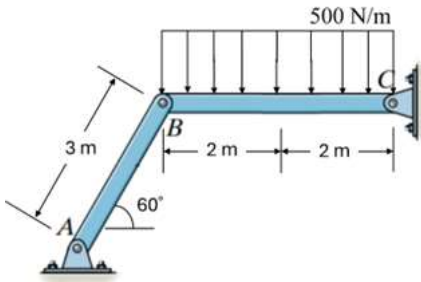
Answer:

$$\begin{cases} AE = 3.4 \text{ kN [C]} \\ AB = BC = 4.5 \text{ kN [T]} \\ BE = 0 \\ BD = 0 \\ CD = DE = 5.2 \text{ kN [C]} \end{cases}$$

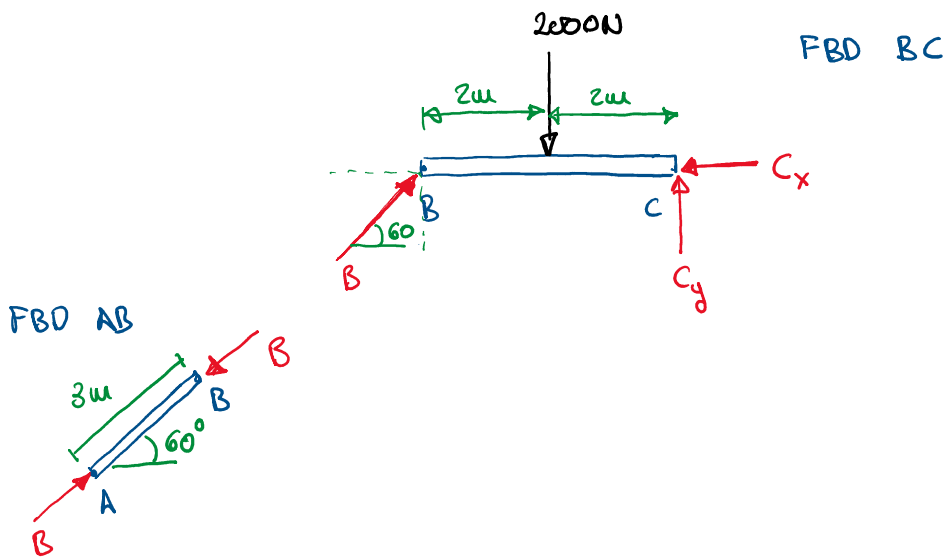
Exercise 4 (30%)

For the frame shown in the figure:

- Draw the Free-Body diagram (FBD) of AB and BC
- Determine the reactions at pin A, B, and C.



a)



- b) Reactions at A, B, C $\Rightarrow$ A = B $\Rightarrow$ solve FBD BC $\rightarrow$ 3 unknown $\left. \begin{array}{l} 2 \text{ known} \end{array} \right\} \checkmark$

$$\sum \mathcal{M}_C = 0 \Rightarrow -4\text{m} \cdot B \sin 60 + 2\text{m} \cdot 2000\text{N} = 0 \Rightarrow \underline{B = 1154.7\text{ N}}$$

$$\sum F_x = 0 \Rightarrow B \cos 60 - C_x = 0 \Rightarrow C_x = 1154.7 \cos 60 \Rightarrow \underline{C_x = 577.35\text{ N}}$$

$$\sum F_y = 0 \Rightarrow B \sin 60 - 2000 + C_y = 0 \Rightarrow C_y = 2000 - (1154.7) \sin 60 \Rightarrow \underline{C_y = 1000\text{ N}}$$

$$C = \sqrt{C_x^2 + C_y^2} = \sqrt{1000^2 + 577.35^2} \Rightarrow \underline{C = 1154.7\text{ N}}$$